ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 144°	One part $= 360 \div 10 = 36^\circ$
2	**(a)** 30	Opposite angles are equal, so the two expressions are equal
3	**(d)** 35°	A diagonal of a rhombus bisects the angle at the vertex
4	**(b)** 50°	OA = OB, so triangle OAB is isosceles
5	**(b)** 110°	Angle sum is 360°
6	**(c)** 115°	Angles on the same non-parallel side add up to 180°
7	**(b)** The diagonals bisect each other	Parallelogram diagonals bisect each other
8	**(d)** 144°	One part $= 360 \div 15 = 24^\circ$
9	**(c)** 15	Opposite angles are equal, so the two expressions are equal
10	**(a)** 50°	A diagonal of a rhombus bisects the angle at the vertex

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total parts 12, one part $= 360 \div 12 = 30^\circ$ (1); smallest angle $= 30^\circ$ (1).	1+1
12	Adjacent angles are supplementary: $3x + 10 + 2x - 10 = 180$ gives $x = 36$ (1); $\angle A = 118^\circ$ (1).	1+1
13	Sum $= 360^\circ$ (1); it equals 360°, so it is possible (1).	1+1
14	Total parts 12, one part $= 360 \div 12 = 30^\circ$ (1); smallest angle $= 60^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the angles be $x, 2x, 3x, 4x$; sum $= 10x = 360$ (1); $x = 36$ (1); angles 36°, 72°, 108°, 144° (1).	1+1+1
16	Diagonal AC bisects $\angle A$: $\angle OAB = 32^\circ$ (1); $\angle ABC = 180 - 64 = 116^\circ$ (1); $\angle OBA = 58^\circ$ (1).	1+1+1
17	OA = OB so $\angle OAB = (180 - 70) \div 2 = 55^\circ$ (1); $\angle OAD = 90 - 55 = 35^\circ$ (1); $\angle AOD = 180 - 70 = 110^\circ$ (1).	1+1+1
	OR: Let the angles be $2x, 3x, 4x, 6x$; sum $= 15x = 360$ (1); $x = 24$ (1); angles 48°, 72°, 96°, 144° (1).	1+1+1
18	Diagonal AC bisects $\angle A$: $\angle OAB = 40^\circ$ (1); $\angle ABC = 180 - 80 = 100^\circ$ (1); $\angle OBA = 50^\circ$ (1).	1+1+1
	OR: OA = OB so $\angle OAB = (180 - 100) \div 2 = 40^\circ$ (1); $\angle OAD = 90 - 40 = 50^\circ$ (1); $\angle AOD = 180 - 100 = 80^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Draw a diagonal; it divides the quadrilateral into two triangles, each with angle sum 180° (3). (ii) $360 - (78 + 92 + 104) = 86^\circ$ (2).	3+2
	OR: (i) They bisect each other (1) at right angles (1). (ii) $\angle OAB = 28^\circ$ (1); $\angle OBA = 90 - 28 = 62^\circ$ (1); $\angle ABC = 124^\circ$ (1).	1+1+1+1+1
20	(i) Draw a diagonal; it divides the quadrilateral into two triangles, each with angle sum 180° (3). (ii) $360 - (66 + 114 + 87) = 93^\circ$ (2).	3+2
	OR: (i) They bisect each other (1) at right angles (1). (ii) $\angle OAB = 35^\circ$ (1); $\angle OBA = 90 - 35 = 55^\circ$ (1); $\angle ABC = 110^\circ$ (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).
- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 40°	OA = OB, so triangle OAB is isosceles
2	**(d)** 100°	Angle sum is 360°
3	**(a)** 108°	Angles on the same non-parallel side add up to 180°
4	**(c)** congruent right triangles	Each triangle has a right angle
5	**(a)** 120°	One part $= 360 \div 12 = 30^\circ$
6	**(b)** 10	Opposite angles are equal, so the two expressions are equal
7	**(c)** 32°	A diagonal of a rhombus bisects the angle at the vertex
8	**(c)** 58°	OA = OB, so triangle OAB is isosceles
9	**(a)** 110°	Angle sum is 360°
10	**(c)** 100°	Angles on the same non-parallel side add up to 180°

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Adjacent angles are supplementary: $2x + 20 + 3x - 10 = 180$ gives $x = 34$ (1); $\angle A = 88^\circ$ (1).	1+1
12	Sum $= 350^\circ$ (1); it is not 360°, so it is not possible (1).	1+1
13	Total parts 20, one part $= 360 \div 20 = 18^\circ$ (1); smallest angle $= 36^\circ$ (1).	1+1
14	Adjacent angles are supplementary: $4x + 5 + 2x + 25 = 180$ gives $x = 25$ (1); $\angle A = 105^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the angles be $3x, 4x, 5x, 6x$; sum $= 18x = 360$ (1); $x = 20$ (1); angles 60°, 80°, 100°, 120° (1).	1+1+1
16	Diagonal AC bisects $\angle A$: $\angle OAB = 26^\circ$ (1); $\angle ABC = 180 - 52 = 128^\circ$ (1); $\angle OBA = 64^\circ$ (1).	1+1+1
17	OA = OB so $\angle OAB = (180 - 64) \div 2 = 58^\circ$ (1); $\angle OAD = 90 - 58 = 32^\circ$ (1); $\angle AOD = 180 - 64 = 116^\circ$ (1).	1+1+1
	OR: Let the angles be $x, 3x, 4x, 4x$; sum $= 12x = 360$ (1); $x = 30$ (1); angles 30°, 90°, 120°, 120° (1).	1+1+1
18	Diagonal AC bisects $\angle A$: $\angle OAB = 55^\circ$ (1); $\angle ABC = 180 - 110 = 70^\circ$ (1); $\angle OBA = 35^\circ$ (1).	1+1+1
	OR: OA = OB so $\angle OAB = (180 - 80) \div 2 = 50^\circ$ (1); $\angle OAD = 90 - 50 = 40^\circ$ (1); $\angle AOD = 180 - 80 = 100^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Draw a diagonal; it divides the quadrilateral into two triangles, each with angle sum 180° (3). (ii) $360 - (120 + 85 + 55) = 100^\circ$ (2).	3+2
	OR: (i) They bisect each other (1) at right angles (1). (ii) $\angle OAB = 42^\circ$ (1); $\angle OBA = 90 - 42 = 48^\circ$ (1); $\angle ABC = 96^\circ$ (1).	1+1+1+1+1
20	(i) Draw a diagonal; it divides the quadrilateral into two triangles, each with angle sum 180° (3). (ii) $360 - (93 + 77 + 101) = 89^\circ$ (2).	3+2
	OR: (i) They bisect each other (1) at right angles (1). (ii) $\angle OAB = 48^\circ$ (1); $\angle OBA = 90 - 48 = 42^\circ$ (1); $\angle ABC = 84^\circ$ (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).
- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** bisect each other at right angles	Rhombus property
2	**(a)** 120°	One part $= 360 \div 18 = 20^\circ$
3	**(c)** 25	Opposite angles are equal, so the two expressions are equal
4	**(d)** 55°	A diagonal of a rhombus bisects the angle at the vertex
5	**(d)** 35°	OA = OB, so triangle OAB is isosceles
6	**(b)** 90°	Angle sum is 360°
7	**(a)** 82°	Angles on the same non-parallel side add up to 180°
8	**(a)** kite	In a kite the axis diagonal bisects the other
9	**(d)** 150°	One part $= 360 \div 12 = 30^\circ$
10	**(c)** 10	Opposite angles are equal, so the two expressions are equal

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $= 360^\circ$ (1); it equals 360°, so it is possible (1).	1+1
12	Total parts 12, one part $= 360 \div 12 = 30^\circ$ (1); smallest angle $= 30^\circ$ (1).	1+1
13	Adjacent angles are supplementary: $5x + 10 + 4x - 10 = 180$ gives $x = 20$ (1); $\angle A = 110^\circ$ (1).	1+1
14	Sum $= 370^\circ$ (1); it is not 360°, so it is not possible (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the angles be $2x, 3x, 5x, 8x$; sum $= 18x = 360$ (1); $x = 20$ (1); angles $40^\circ, 60^\circ, 100^\circ, 160^\circ$ (1).	1+1+1
16	Diagonal AC bisects $\angle A$: $\angle OAB = 38^\circ$ (1); $\angle ABC = 180 - 76 = 104^\circ$ (1); $\angle OBA = 52^\circ$ (1).	1+1+1
17	OA = OB so $\angle OAB = (180 - 110) \div 2 = 35^\circ$ (1); $\angle OAD = 90 - 35 = 55^\circ$ (1); $\angle AOD = 180 - 110 = 70^\circ$ (1).	1+1+1
	OR: Let the angles be $4x, 5x, 6x, 9x$; sum $= 24x = 360$ (1); $x = 15$ (1); angles $60^\circ, 75^\circ, 90^\circ, 135^\circ$ (1).	1+1+1
18	Diagonal AC bisects $\angle A$: $\angle OAB = 47^\circ$ (1); $\angle ABC = 180 - 94 = 86^\circ$ (1); $\angle OBA = 43^\circ$ (1).	1+1+1
	OR: OA = OB so $\angle OAB = (180 - 50) \div 2 = 65^\circ$ (1); $\angle OAD = 90 - 65 = 25^\circ$ (1); $\angle AOD = 180 - 50 = 130^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Draw a diagonal; it divides the quadrilateral into two triangles, each with angle sum 180° (3). (ii) $360 - (84 + 96 + 112) = 68^\circ$ (2).	3+2
	OR: (i) They bisect each other (1) at right angles (1). (ii) $\angle OAB = 60^\circ$ (1); $\angle OBA = 90 - 60 = 30^\circ$ (1); $\angle ABC = 60^\circ$ (1).	1+1+1+1+1
20	(i) Draw a diagonal; it divides the quadrilateral into two triangles, each with angle sum 180° (3). (ii) $360 - (105 + 65 + 98) = 92^\circ$ (2).	3+2
	OR: (i) They bisect each other (1) at right angles (1). (ii) $\angle OAB = 24^\circ$ (1); $\angle OBA = 90 - 24 = 66^\circ$ (1); $\angle ABC = 132^\circ$ (1).	1+1+1+1+1

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- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).
- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

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